

# Stochastic Series Expansion Quantum Monte Carlo for Spin- $\frac{1}{2}$ Systems on Bipartite Lattices

A Pedagogical Companion to `qmc_sse`

The `qmc_sse` contributors

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## Abstract

We give a self-contained, pedagogical exposition of the *Stochastic Series Expansion* (SSE) Quantum Monte Carlo algorithm with operator-loop cluster updates, as introduced by Sandvik [2] for the spin- $\frac{1}{2}$  antiferromagnetic Heisenberg model on bipartite lattices. Starting from the bare high-temperature expansion of the partition function, we derive the SSE configuration weight, sketch the sign-problem-free Marshall sign transformation, and lay out the diagonal Metropolis update, the operator-loop off-diagonal update, and the standard estimators for thermodynamic and magnetic observables. We close with a practical map onto the `qmc_sse` implementation and a short verification table.

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## 1 Introduction

Quantum Monte Carlo (QMC) is the workhorse for unbiased simulations of strongly correlated quantum many-body systems. For spin systems on *bipartite* lattices, the Stochastic Series Expansion (SSE) of Sandvik and Kurkijärvi [1], in its modern *operator-loop* formulation [2], is the method of choice. It has near-perfect ergodicity, very short autocorrelation times even close to phase transitions, no Trotter error, and only modest memory overhead.

This note is an applied tutorial. The mathematical level is graduate statistical physics; the only prerequisites are spin- $\frac{1}{2}$  operators, the Boltzmann distribution, and a working knowledge of Markov-chain Monte Carlo. References to the `qmc_sse` C++ codebase [7] are interleaved with the theory so the reader can connect each formula to the actual implementation.

## 2 Model and notation

We consider the spin- $\frac{1}{2}$  antiferromagnetic (AF) Heisenberg Hamiltonian on a  $d$ -dimensional bipartite lattice with  $N_s$  sites and  $N_b$  nearest-neighbour bonds:

$$\hat{H} = J \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j = J \sum_{b=1}^{N_b} \hat{H}_b, \quad J > 0, \quad (1)$$

where the bond operator

$$\hat{H}_b = \hat{S}_i^z \hat{S}_j^z + \frac{1}{2} (\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+) \quad (2)$$

acts on the two sites  $(i, j)$  joined by bond  $b$ . We work in the local  $\hat{S}^z$  basis where  $\hat{S}^z |\sigma\rangle = \frac{\sigma}{2} |\sigma\rangle$ ,  $\sigma \in \{-1, +1\}$ . We denote a many-body basis state by  $|\alpha\rangle = |\sigma_1 \sigma_2 \dots \sigma_{N_s}\rangle$ .

**Bipartite lattice.** By bipartite we mean the sites split into two sublattices  $A, B$  such that every bond connects a site in  $A$  to one in  $B$ . Equivalently,  $(-1)^{\text{sub}(i)+\text{sub}(j)} = -1$  on every bond. The chain (even length), square, and honeycomb lattices satisfy this; triangular and kagome lattices do not. In `qmc_sse` the helper `Lattice::is_bipartite()` performs this check at construction time.

## 3 Series expansion of the partition function

The starting point is the high-temperature series expansion of the canonical partition function:

$$Z = \text{Tr} e^{-\beta \hat{H}} = \sum_{n=0}^{\infty} \frac{(-\beta)^n}{n!} \text{Tr} \hat{H}^n. \quad (3)$$

Inserting eq. (1) and resolving each  $\hat{H}^n$  as a sum over ordered strings of bond labels gives

$$Z = \sum_{\alpha} \sum_{n=0}^{\infty} \sum_{S_n} \frac{(-\beta)^n J^n}{n!} \langle \alpha | \hat{H}_{b_n} \hat{H}_{b_{n-1}} \dots \hat{H}_{b_1} | \alpha \rangle, \quad (4)$$

where  $S_n = (b_1, b_2, \dots, b_n)$  runs over all ordered bond sequences of length  $n$ .

**Decomposition into diagonal and off-diagonal parts.** For the loop algorithm to be efficient we split each bond operator into a positive-definite *diagonal* and a positive-definite *off-diagonal* piece. Define

$$H_{1,b} = \frac{1}{4} - \hat{S}_i^z \hat{S}_j^z, \quad (5)$$

$$H_{2,b} = \frac{1}{2} (\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+). \quad (6)$$

Both have matrix element  $\frac{1}{2}$  on antiparallel-spin bond states  $|\uparrow\downarrow\rangle$  or  $|\downarrow\uparrow\rangle$ , and *zero* otherwise:

$$\langle\uparrow\downarrow|H_{1,b}|\uparrow\downarrow\rangle = \langle\downarrow\uparrow|H_{1,b}|\downarrow\uparrow\rangle = \frac{1}{2}, \quad \langle\uparrow\downarrow|H_{2,b}|\downarrow\uparrow\rangle = \langle\downarrow\uparrow|H_{2,b}|\uparrow\downarrow\rangle = \frac{1}{2}. \quad (7)$$

In terms of  $H_{1,b}, H_{2,b}$  the bond Hamiltonian reads

$$\hat{H}_b = \frac{1}{4} - H_{1,b} - H_{2,b}^{\text{sign}}, \quad (8)$$

where  $H_{2,b}^{\text{sign}}$  is the off-diagonal piece *with its physical sign*. To use this decomposition without sign problems we must absorb that sign—which brings us to the Marshall transformation.

### 3.1 Marshall sign transformation

Let  $U = \prod_{i \in B} e^{i\pi \hat{S}_i^z}$  be the unitary that rotates all  $B$ -sublattice spins by  $\pi$  around  $\hat{z}$ . It acts as the identity on  $\hat{S}_i^z$  and flips the sign of  $\hat{S}_i^+, \hat{S}_i^-$  on  $B$  sites. On a bipartite lattice every bond couples one  $A$  site to one  $B$  site, so each off-diagonal term picks up a minus sign:

$$\tilde{H} \equiv U^\dagger \hat{H} U = J \sum_b \left[ \hat{S}_i^z \hat{S}_j^z - \frac{1}{2} (\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+) \right]. \quad (9)$$

Identifying eq. (9) with the  $H_{1,b}, H_{2,b}$  basis,

$$\boxed{\tilde{H} = -J \sum_{b=1}^{N_b} (H_{1,b} + H_{2,b}) + \frac{J N_b}{4} \hat{\mu}^z.} \quad (10)$$

Both  $H_{1,b}$  and  $H_{2,b}$  now appear with a *negative* prefactor, so the SSE weights below come out positive. This is the celebrated bipartite-lattice “cure” of the sign problem; it does *not* extend to frustrated geometries.

### 3.2 Truncated, padded operator string

Substituting eq. (10) into eq. (3) and truncating the sum at a large fixed length  $M$ , one obtains

$$Z = \sum_{\alpha} \sum_{S_M} \beta^n \frac{(M-n)!}{M!} \prod_{p=1}^M \langle \alpha_p | J H_{a_p, b_p} | \alpha_{p-1} \rangle, \quad (11)$$

where  $S_M = \{(a_1, b_1), \dots, (a_M, b_M)\}$  is now a sequence of length  $M$ ,  $a_p \in \{0, 1, 2\}$  encodes (identity, diagonal, off-diagonal),  $b_p$  is the bond label for non-identity entries,  $n$  is the number of non-identity entries in  $S_M$ , and the intermediate states  $|\alpha_p\rangle$  are obtained by propagating  $|\alpha_0\rangle \equiv |\alpha\rangle$  forward through the operator string. The combinatorial factor  $(M-n)!/M!$  accounts for the  $\binom{M}{n}$  ways of placing  $n$  non-identity operators in a length- $M$  slot, with identities  $H_{0,b} \equiv \hat{\mu}^z$  filling the gaps. Choosing

$$M \gtrsim 1.4 \langle n \rangle \quad (12)$$

suffices to make the truncation error negligible. In `qmc_sse` the cutoff is grown adaptively during thermalization in `OperatorString::grow_if_needed()`.

**Configuration weight.** The SSE Markov chain samples configurations  $\mathcal{C} = (\alpha, S_M)$  with weight

$$W(\mathcal{C}) = \beta^n \frac{(M-n)!}{M!} \prod_{p=1}^M \langle \alpha_p | J H_{a_p, b_p} | \alpha_{p-1} \rangle. \quad (13)$$

On a bipartite lattice  $W(\mathcal{C}) \geq 0$ , and the matrix elements are  $1/2$  for every non-identity operator, so each non-identity operator contributes a multiplicative factor  $J/2$  to  $W$ .

## 4 Monte Carlo updates

A single Monte Carlo step in `qmc_sse` consists of

- (i) a *diagonal update* (Metropolis sweep through  $S_M$ ),
- (ii) an *operator-loop update* on the linked-vertex list (the off-diagonal cluster move), and
- (iii) a *free-spin flip* for sites unvisited by any operator.

### 4.1 Diagonal update

Sweep  $p = 1, \dots, M$ . The local move at slot  $p$  depends on its current contents:

**Identity  $\rightarrow$  diagonal insertion.** Pick a random bond  $b$ . If the spins on  $b$  are antiparallel in the current propagated state  $|\alpha_p\rangle$ , attempt insertion of  $H_{1,b}$  at slot  $p$ . The Metropolis acceptance ratio is the ratio of weights of the proposed and current strings,

$$P_{\text{ins}} = \min\left(1, \frac{\beta N_b J/2}{M-n}\right). \quad (14)$$

The factor  $N_b$  comes from the inverse selection probability of the bond, and  $(M-n)$  counts the number of identity slots that could have hosted the insertion.

**Diagonal  $\rightarrow$  identity removal.** Conversely,

$$P_{\text{rem}} = \min\left(1, \frac{M-n+1}{\beta N_b J/2}\right). \quad (15)$$

By construction  $P_{\text{ins}} P_{\text{rem}}^{-1}$  satisfies detailed balance for the SSE weight eq. (13).

**Off-diagonal slot.** If  $a_p = 2$  the slot is left untouched in the diagonal sweep, but  $|\alpha_p\rangle$  is propagated forward by exchanging the two spins on bond  $b_p$ .

The diagonal sweep alone is not ergodic: it cannot change the *number* of off-diagonal operators. The operator-loop update takes care of that.

### 4.2 Linked-vertex list

We rebuild the linked-vertex list from scratch before every loop sweep. Each non-identity operator at slot  $p$  is a *vertex* with four legs (fig. 1):

$$\begin{aligned} \ell_0 &= \text{bottom-left (site } i, \text{ in)}, & \ell_1 &= \text{bottom-right (site } j, \text{ in)}, \\ \ell_2 &= \text{top-left (site } i, \text{ out)}, & \ell_3 &= \text{top-right (site } j, \text{ out)}. \end{aligned}$$

Sweeping operators in order, we link each leg to the previous unmatched leg on the same site; the very first and last legs on each site are joined cyclically across the periodic time direction. The result is a 1D doubly-linked structure indexed by global leg id  $4v + k$ ,  $k \in \{0, 1, 2, 3\}$ . See `LinkedVertices::build` in `include/qmc/operator_string.hpp`.

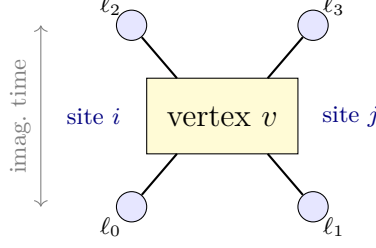


Figure 1: The four legs of an SSE vertex. Bottom legs ( $\ell_0, \ell_1$ ) hold the spins *before* the operator acts; top legs ( $\ell_2, \ell_3$ ) hold them *after*.

### 4.3 Operator-loop update

For the AF Heisenberg model in the  $\hat{S}^z$  basis the loop traversal rule is the deterministic *switch*:

$$\text{enter leg } \ell \implies \text{exit leg } \ell \oplus 1, \quad (16)$$

i.e. stay on the same time-side (top or bottom) but jump to the *other* site of the bond. From the exit leg we follow the linked-vertex list pointer to the next leg, and the loop continues until it closes on its starting leg.

A loop visits exactly two legs per vertex (one on each side it touches). After all loops have been constructed, each independent loop is flipped with probability 1/2. The bookkeeping for the operator string is:

- if a vertex has *exactly one* of its time-sides touched by a flipped loop, its diagonal/off-diagonal type is toggled;
- if both sides are touched (or neither), the type is left unchanged;
- the persistent spin state  $|\alpha\rangle \equiv |\alpha_0\rangle$  on each site  $i$  is flipped iff the very first leg encountered for site  $i$  belongs to a flipped loop.

The proof that this leaves the SSE weight eq. (13) invariant is elementary: every flipped loop simultaneously flips two spins per visited vertex on the touched time-side, mapping antiparallel pairs to antiparallel pairs (still nonzero matrix element) and exchanging the diagonal  $\leftrightarrow$  off-diagonal labels in lock-step with the spin pattern.

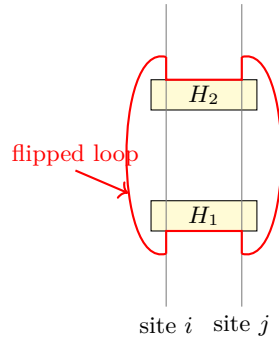


Figure 2: An operator-loop drawn on two stacked vertices acting on the same bond. The loop touches the bottom side of both vertices (equivalently, the bottom legs  $\ell_0$  and  $\ell_1$  of each). Flipping the loop toggles the spins on those legs and exchanges the diagonal/off-diagonal type of both vertices.

#### 4.4 Free-spin flip

Sites with no operators acting on them throughout the operator string live on a free worldline whose spin state is unconstrained. Such a spin is flipped with probability  $1/2$ . This is essential at small  $\beta$  or low operator densities where many sites would otherwise remain stuck in their initial random orientation.

### 5 Adaptive truncation

The truncation  $M$  should comfortably exceed any  $n$  that the simulation visits. During thermalization we monitor  $n_{\max}$  and grow  $M$  to

$$M \leftarrow \max(M, \lceil 1.4 n_{\max} \rceil + 4). \quad (17)$$

Once measurement starts  $M$  is frozen so that the weight normalization is constant across samples.

### 6 Estimators

A central virtue of SSE is that many key observables are *operator-string* estimators: they only require knowledge of  $n$  or simple sums over the propagated states.

#### 6.1 Energy and specific heat

Differentiating  $\ln Z$  with respect to  $\beta$  in eq. (11) gives (Sandvik, 2010 [4], Sec. 5):

$$E = -\frac{\langle n \rangle}{\beta} + \frac{J N_b}{4}, \quad (18)$$

$$C_v = \langle n^2 \rangle - \langle n \rangle^2 - \langle n \rangle. \quad (19)$$

The first equation holds per configuration; the second requires the fluctuation  $\langle n^2 \rangle - \langle n \rangle^2$  (and is best accompanied by a delete-one jackknife to propagate errors through the nonlinear combination [5]). In `qmc_sse` both estimators live in `Measurements`.

#### 6.2 Magnetic observables

Conserved  $\hat{S}_{\text{tot}}^z$  on each operator-loop step means that the  $z$  component of the total magnetization at the  $p = 0$  slice is already a faithful sample of the canonical distribution. Define

$$m_z = \frac{1}{2N_s} \sum_i s_i, \quad m_{\text{stag}} = \frac{1}{2N_s} \sum_i (-1)^{\text{sub}(i)} s_i, \quad (20)$$

in units where  $\hat{S}^z = s/2$ . The simulation accumulates  $\langle |m_z| \rangle, \langle m_z^2 \rangle, \langle |m_{\text{stag}}| \rangle, \langle m_{\text{stag}}^2 \rangle, \langle m_{\text{stag}}^4 \rangle$ , plus the static susceptibilities

$$\chi_q = \frac{\beta}{N_s} \langle M_q^2 \rangle, \quad M_q = \sum_i e^{i\mathbf{q} \cdot \mathbf{r}_i} \hat{S}_i^z, \quad (21)$$

specialized to  $q = 0$  (uniform) and  $q = \pi$  (staggered).

#### 6.3 Binder cumulant

The fourth-order cumulant of the staggered magnetization,

$$U_4 = 1 - \frac{\langle m_{\text{stag}}^4 \rangle}{3 \langle m_{\text{stag}}^2 \rangle^2}, \quad (22)$$

is the canonical scaling function for locating phase transitions: at the critical point  $U_4$  is system-size independent, while it tends to  $\frac{2}{3}$  in the ordered phase and to 0 in the paramagnet.

## 7 Error analysis

QMC samples are correlated, so naive standard errors underestimate the true uncertainty. We use the Flyvbjerg–Petersen logarithmic binning analysis [5]: incoming samples are pushed into bin level 0, every two adjacent bin-level- $k$  samples are averaged and propagated to bin level  $k + 1$ , and so on. A plateau of the standard error as a function of bin level signals that the bin size has surpassed the integrated autocorrelation time  $\tau_{\text{int}}$ . The implementation is in `qmc::Observable`.

For derived observables like  $C_v$  that mix two means in a nonlinear way, the codebase additionally uses a delete-one jackknife on the raw samples (`qmc::JackknifeSamples`).

## 8 Map to the codebase

The following table connects every formula in this note to its implementation site:

| Object                   | Formula     | Implementation                                  |
|--------------------------|-------------|-------------------------------------------------|
| SSE configuration weight | eq. (13)    | <code>SseEngine::diagonal_update_</code>        |
| Insertion acceptance     | eq. (14)    | <code>prefactor_insert_base / (M-n)</code>      |
| Removal acceptance       | eq. (15)    | <code>(M-n+1) / prefactor_insert_base</code>    |
| Linked-vertex list       | section 4.2 | <code>LinkedVertices::build</code>              |
| Switch rule              | eq. (16)    | <code>exit_leg = leg ^ 1</code>                 |
| Adaptive $M$             | eq. (17)    | <code>OperatorString::grow_if_needed</code>     |
| Energy estimator         | eq. (18)    | <code>Measurements::measure</code>              |
| Specific heat            | eq. (19)    | <code>Measurements::specific_heat</code>        |
| Susceptibilities         | eq. (21)    | <code>Measurements::measure</code>              |
| Binder cumulant          | eq. (22)    | <code>Measurements::binder_cumulant</code>      |
| Error analysis           | section 7   | <code>qmc::Observable / JackknifeSamples</code> |

**Reproducing the verification table.** The unit tests `tests/test_heisenberg_chain.cpp` recover the key analytic limits to the precision allowed by their (modest) sample counts:

| System              | $\beta$ | Quantity            | Reference                                | <code>qmc_sse</code> |
|---------------------|---------|---------------------|------------------------------------------|----------------------|
| Chain $L = 4$ ring  | 8       | $E/N_s$             | $-0.500$ (exact)                         | $-0.498 \pm 0.001$   |
| Chain $L = 8$       | 0.05    | $\langle n \rangle$ | $\beta J N_b / 4 = 0.10$ (lead.)         | $0.099 \pm 0.001$    |
| Chain $L = 8$       | 0.05    | $E/N_s$             | $\rightarrow 0$ as $\beta \rightarrow 0$ | $\sim 10^{-3}$       |
| Square $8 \times 8$ | 8       | $E/N_s$             | $\sim -0.669$ (g.s. lit.)                | $-0.6720 \pm 0.0005$ |

## 9 Computational complexity

Per Monte Carlo step the dominant costs are:

- Diagonal sweep:  $\mathcal{O}(M + N_s)$ .
- Linked-vertex build:  $\mathcal{O}(n + N_s)$ .
- Loop traversal:  $\mathcal{O}(n)$  amortized (each leg visited at most once).
- Free-spin flip:  $\mathcal{O}(N_s)$ .

For the AF Heisenberg model  $\langle n \rangle \sim \beta J N_b / 4$  at high temperature and grows roughly as  $\beta J N_b \times c$  with a model-dependent constant  $c < 1$  at low temperature; memory is  $\mathcal{O}(M + 4\langle n \rangle)$  words.

## 10 Limitations and outlook

The implementation in this note (and the accompanying code) has a deliberately narrow scope. Three natural extensions are:

- *Other models.* The directed-loop equations of Syljuåsen and Sandvik [3] subsume the switch rule used here and apply to a wide class of bilinear spin Hamiltonians (e.g. XXZ, transverse-field Ising, bilinear-biquadratic). Adding a new model boils down to coding its bond vertex table and the corresponding directed-loop probabilities.
- *Frustrated systems.* The Marshall sign is unique to bipartite lattices. Frustrated geometries (triangular, kagome,  $J_1$ - $J_2$ ) exhibit a sign problem that is, in general, NP-hard [6] and lies outside the scope of vanilla SSE.
- *Replica exchange.* The current code parallelises *independent* replicas at the same  $(J, \beta)$ . Adding parallel tempering across  $\beta$  values would help equilibrate at low temperatures.

## A Reference: pseudocode of one MC step

Listing 1: Skeleton of `SseEngine::mc_step()`.

```
1 void SseEngine::mc_step() {
2     diagonal_update_(); // Metropolis sweep over the operator string
3     loop_update_();     // operator-loop cluster move (off-diagonal)
4     flip_free_spins_(); // independent flips of unvisited sites
5 }
6
7 // Diagonal sweep:  $O(M + N_s)$  per call.
8 void SseEngine::diagonal_update_() {
9     auto state = spins_; // propagated state
10    const Real prefactor = beta_ * Nb * J/2; // see eq. (P_ins)
11    for (Length p = 0; p < M; ++p) {
12        OpCode op = ops_[p];
13        if (op_is_identity(op)) {
14            Bond b = rng_.uniform_int(Nb);
15            const auto& bd = lat_.bond(b);
16            if (state[bd.i] != state[bd.j]) {
17                Real Pacc = prefactor / (M - n_op());
18                if (rng_.uniform() < Pacc) ops_.set(p, pack_op(b, 0));
19            }
20        } else if (op_is_diagonal(op)) {
21            Real Pacc = (M - n_op() + 1) / prefactor;
22            if (rng_.uniform() < Pacc) ops_.set(p, kIdentity);
23        } else { // off-diagonal
24            const auto& bd = lat_.bond(op_bond(op));
25            state[bd.i] = -state[bd.i];
26            state[bd.j] = -state[bd.j];
27        }
28    }
29 }
```

The full implementation, including the linked-vertex builder and the loop traversal, lives in `include/qmc/sse_engine.hpp` and is annotated with line-by-line cross-references to the equations of this note.

## References

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